Cup Holder - Tea Reminder

Iliasă Antonia
group 30433
Contact: Iliasa.Ma.Antonia@student.utcluj.ro

1. Introduction

The Smart Cup Holder weighs the cup sitting on it and uses a buzzer to remind the user to drink. When you work at a desk, it's easy to forget your tea until it has gone cold. This project fixes that with a simple weight sensor and a sound buzzer. A cup is considered present when the measured mass exceeds a 50 g threshold for at least 300 ms. The system plays a short melody when the cup is lifted and a repeating alarm when the cup stays untouched for 5 minutes.

The holder does two things:

- **It confirms each sip.** Internally it remembers every time the cup is lifted off the holder.
- **It reminds you to drink.** If the cup sits untouched for a set time (5 minutes by default), the buzzer beeps repeatedly until the cup is picked up.

2. Components

The whole build needs the following parts and a USB cable:

Component	Quantity	Usage
1 kg Load cell	1	Senses the weight of the cup; its resistance changes slightly as it bends
HX711 module	1	24-bit amplifier and converter that turns the tiny load cell signal into a number the Arduino can read

Arduino Mega	1	Reads the weight, decides when the cup is lifted or forgotten, drives the buzzer
Passive buzzer	1	Plays the lift beep and the reminder alarm
Cup holder base + top plate		Used two plates of wood to separate the load cell
Jumper wires		Connect the modules
USB cable / 5 V supply		Powers the Arduino

3. Theoretical background

The measurement chain converts the mass of the cup into a digital value in three stages: mechanical strain in the load cell, a differential voltage across a Wheatstone bridge, and analog-to-digital conversion in the HX711. A load cell is an aluminium beam with strain gauges bonded to it. A mass m placed on the beam bends it and produces a strain ε. The relative change in resistance of each gauge is proportional to this strain:

$$\frac{\Delta R}{R} = GF \cdot \varepsilon$$

where GF is the gauge factor, approximately 2 for metal-foil gauges. The four gauges are connected as a full Wheatstone bridge, whose output voltage is proportional to the applied mass:

$$Vout = S \cdot Vexc \cdot \frac{m}{Mmax}$$

for S is the sensitivity, Vexc the excitation voltage and Mmax = 1 kg the rated capacity. For a typical 1 kg cell with S = 1 mV/V and V_exc = 5 V the full-scale output is 1 mV/V × 5 V = 5 mV at 1 kg, and the sensitivity is therefore about 5 μV per gram.

In order to convert such a small signal to the actual voltage the arduino board uses, about 4.9mV for its built in 10-bit ADC (5V / 1024) we use the HX711 module to amplify the signal. The HX711 amplifies the signal with a programmable gain of 128 and digitises it with a 24-bit sigma-delta converter. At gain 128 and a 5 V supply its full-scale differential input range is ±20 mV, so the 5 mV full-scale signal uses about a quarter of that range. The output data rate is 10 or 80 samples per second.

4. How it works

Since a 1kg load-cell was used, the cup itself with the liquid has to be less than 1kg to be correctly measured. The holder knows whether the cup is on it by checking if the measured weight is above a threshold (50 g). A load cell is a metal bar with strain gauges glued on it. When weight is placed on the bar, it bends very slightly and the gauges' resistance changes. The gauges form a Wheatstone bridge, so the change appears as a tiny voltage, only a few millivolts.

The HX711 module. That voltage is too small for the Arduino to read directly. The HX711 amplifies it and converts it into a 24-bit digital value, then sends it to the Arduino over two wires (DT for data, SCK for clock).

The Arduino. The Arduino converts the raw value into grams using a calibration factor. It then compares the weight to the threshold:

- Weight above 50 g : the cup is on the holder.
- Weight below 50 g : the cup has been lifted.

A change only counts if it lasts at least 300 ms. This stops a bump on the desk from triggering a false beep. Because an empty cup still weighs well over 50 g, the holder works whether the cup is full or nearly empty.

5. Proposed improvements

As future work a temperature-controlled heating can be added. At present the drink still cools if the reminder is ignored. A temperature sensor pressed against the bottom of the cup, such as a DS18B20 probe or an NTC thermistor, would measure the drink temperature. A resistive heating pad, switched by the Arduino through a MOSFET, would heat the drink only when its temperature falls below a set point. A two-point (hysteresis) controller would keep it in a drinking range instead of boiling it, for example switching the heater on below 55 °C and off above 60 °C. Because the load cell already detects the cup, the heater would switch off whenever the cup is lifted or the holder is empty. In the planned sequence, heating starts if the cup has still not been lifted after the 2-minute escalating alarm.

The main risk is a tea bag left in the cup. Because the drink stays well below 100 °C, the bag cannot burn, but keeping the tea hot for a long time keeps extracting tannins and makes it bitter. This would be limited by a maximum heating time (for example 30 min) and by advising the user to remove the bag before the cup is placed on the holder. The heater would also need its own power supply, a thermal fuse as a safety cut-off, and insulation from the load cell, whose readings drift with temperature.